



JECRCTM
UNIVERSITY
BUILD YOUR WORLD

Professional Skills

Unit-3

Cognitive and Non-Cognitive Skills



JECRC Foundation

1) Meaning

Cognitive skills are also known as cognitive functions, cognitive abilities or cognitive capacities.

Cognitive skills are essential for human adaptation and survival.

2) Types of Cognitive Skills

Cognitive skills are divided into nine different categories, each reflecting a different method that the brain uses to effectively interpret and use information.

- Sustained attention
- Selective attention
- Divided attention
- Long-term memory
- Working memory
- Logic and reasoning
- Auditory processing
- Visual processing
- Processing speed

3) Strategies to Develop Cognitive Skills

While cognitive skills are not generally amenable to direct instruction, they can be developed through the right kind of cognitive training or brain training. Effective cognitive training incorporates key neuroscience principles to be successful.

Physical and mental exercises: Physical exercises (like walking, exercising, yoga, or playing some physical games) and mental exercises (meditation and being mindful of thoughts) help in developing cognitive skills.

Practice focusing: Purposeful focusing of the mind, removing distractions, and engaging more senses can help in focusing attention. Taking a brief walk around the workspace after having sat for a long duration, or listening to music (using headphones so as not to disturb others) to help re-focus) are simple methods.

Exercising the brain: Activities such as reading a book, playing mind games, writing a story or memorizing a favourite song or poem; hobbies and learning something new (difficult, but achievable) every day stimulate certain areas of the brain and improve the associated cognitive skills and reduce cognitive decline.

Essential Cognitive Skills at the Workplace :-

Cognitive skills are very important in the workplace. These skills help to interpret data, remember team goals, pay focused attention, and more.

Highlighting Cognitive Skills during Job Search :-

Almost all jobs including manual labour require the workers to exercise their cognitive skills. It is important for job seekers to be able to demonstrate that they have the types of cognitive skills employers want.

Critical Thinking Skills



Meaning: Critical thinking is a valuable skill for all aspects of life. It benefits problem- solving, creativity, and teamwork. It is also very much required at the workplace. Critical thinking is a way of approaching whatever is presently occupying the mind to come to the best possible conclusion.

Problem-solving skills

I. Meaning

According to the Concise Oxford Dictionary (1995) a problem is, A doubtful or difficult matter requiring a solution and Something hard to understand or accomplish or deal We are continuously exposed to opportunities in life, at home, school, at work, and even in the community. However, sometimes we miss opportunities or do not take full advantage of them.

Problem Solving Skills



Stages of Problem-solving

Many approaches may be used to solve problems but the following stages are required for a workable solution:

1. Define the problem
2. Brainstorm alternatives
3. Choose the best strategy
4. Implement your solution

Problem Solving Skills



Problem-solving Strategies

People may use many different ways in solving problems. Some of the strategies people employ to figure out and fix a problem are:

- i) **Algorithms:** An algorithm is a step-by-step procedure that always produces the correct solution. For example, a mathematical formula is a good example of a problem-solving algorithm. This strategy, however, is not practical for many situations because it can be time-consuming.
- ii) **Heuristics:** A heuristic is a mental rule-of-thumb strategy that may or may not work in certain situations. Heuristics do not always guarantee correct solutions. However, heuristics do allow people to simplify complex problems and reduce the total number of possible solutions to a more manageable set.
- iii) **Trial and error:** This approach involves trying out several different solutions and ruling out those that do not work in problem-solving. This approach can be appropriate when a very limited number and similarly weighted options are available.
- iv) **Insight:** Sometimes, the solution to a problem appears as a sudden insight. This can occur because the problem is actually similar to something dealt with in the past or is present as an exemplar.

Meaning :-

No matter what you may set your sights on doing or becoming, if you want to be a master the ability to learn that will get you in the game and keep you a contender, there.

Non-cognitive Skills:



Non-cognitive Skills: Meaning, Types of Non-cognitive Skills and Strategies :-

I. Meaning and Types

Non-cognitive skills are defined as the of thought, feelings and (Borghans et al., 2008) that are socially determined and can be developed throughout the lifetime to produce value. Non-cognitive skills comprise personal traits, attitudes, and motivations. Non-cognitive skills have a positive effect on employment outcomes and these can be developed through education and training.

Empathy

I. Meaning

Empathy is the ability to recognize emotions in others, and to understand other perspectives on a situation. Empathy and sympathy are two terms used for such feelings, but there is a difference between empathy and sympathy.

I. Meaning

Vince Lombardi commitment to a group effort that is what makes a team work, a company work, a society work, a civilization .

Andrew Carnegie ability to work together toward a common vision. The ability to direct individual accomplishments toward organizational objectives. It is the fuel that allows common people to attain uncommon

Teamwork



Teamwork is a process in which team members, using their individual knowledge, experience and skills through dynamic interaction with other team members, seek to achieve the common goals of the group/organization. Teamwork focuses on shared behaviours, cognitions, and attitudes of team members. Communication plays an important role in teamwork. It helps team members to resolve disagreements, synchronize information from multiple sources, align toward goals, or distribute critical information to each other.

I. Meaning

Creativity is a mental and social process involving the generation of new ideas or concepts, or new associations of the creative mind between existing ideas or concepts. Another point of view is that it is simply the act of making something new.

Creative Skills at the Workplace

Creativity is also required at the workplace. The following examples of creative skills can be practised to become more creative in the workplace. Used in combination, these can be helpful in thinking differently about a problem or a task:

- Making connections
- Asking questions
- Making observations
- Networking
- Experimenting

I. Meaning

Collaboration is a purposeful relationship in which the concerned parties decide strategically to cooperate to achieve shared or overlapping objectives. Teams that work collaboratively are more likely to obtain greater resources, recognition, and rewards, especially when facing competition for limited resources.

Critical Thinking Skills



Collaborative learning can be between two peers or in small/larger groups. Peer learning is a type of collaborative learning that involves learners working in pairs or in small groups to discuss concepts or find solutions to problems. It helps in addressing misunderstandings and clarifying misconceptions. Research shows that educational experiences that are active, social, contextual, engaging, and learner-owned lead to deeper learning.

I. Meaning

Resilience is the process of adapting well in the face of adversity, trauma, tragedy, threats, or significant stress. The stressors may be family and relationship problems, serious health problems, or related to workplace and financial issues. Emotions of pain and sadness are common in people who have suffered major adversity or trauma in their lives.

Interpersonal skills



Interpersonal skills are sometimes referred to as social skills, people skills, soft skills, or life skills. We need these skills and use them to communicate and interact with other people both individually and in groups.

Variants of Interpersonal Skills:

Communication skills:

Verbal Communication with oral language.

Emotional Intelligence: The ability to understand and manage personal and emotions.

Team Work: Ability to work formally or informally with others in groups and in teams.

Negotiation, Persuasion and Influencing Skills: The ability to work with others to find a mutually agreeable outcome. Although this may be considered a subset of communication, it is often treated separately.

Conflict Resolution: Working with others to resolve interpersonal conflicts and disagreements in a positive way.

Perseverance



Perseverance usually helps an individual to keep trying repeatedly to overcome obstacles or challenges and to accomplish the final goal. Perseverance and grit are malleable by environmental factors. It can be improved by reflecting on failures, belief in the changeability of brain or cognitive abilities, as well as by focusing on envisioning and planning in breaking obstacles.

Self-control

Self-control is defined as an capacity for altering own responses according to the standards of a social group such as ideals, values, morals, and social expectations, and to support the pursuit of long-term goals. Some other terms such as self-regulation, self-discipline, will-power, effortful control, ego strength, and inhibitory control are also used by researchers to describe self-control.

Social skills



Social skills can be defined as the ability to establish a compatible and effective relationship with others. It can be understood as an knowledge of and ability to use appropriate social behaviours that are pleasing to others in interpersonal situations.

Peer Pressure

I. Meaning:

A peer can be any individual who belongs to the same social groups or circles and has some type of influence over other members of the group. Members of a peer group may influence a beliefs and behaviour. A group or individual may be encouraged and want to follow their peers by changing their attitudes, values or behaviours to conform to those of the influencing group or individual.

I. Meaning:

Stress is a mental reaction that our body experiences due to a demanding circumstance or event that requires immediate action and attention. Stress is defined as a state of mental or emotional strain caused by adverse circumstances.

Causes of Stress:

The actual cause of stress varies from person to person. The same situation may be stressful for an individual but challenging for others.